1. What is the primary purpose of a Use Case Model?

A) To depict the internal code structure of the system

B) To graphically represent the proposed functionality and capture functional requirements

C) To design the user interface layout

D) To calculate the project budget and timeline

E) To select the programming language for development

Answer: B) To graphically represent the proposed functionality and capture functional requirements

Explanation: Slide 4 states that the Use Case Model "graphically represent the proposed functionality" and "captures the functional requirements of a system."

2. Use cases are an excellent way to communicate to stakeholders:

A) How the system will be coded

B) The specific algorithms that will be used

C) WHAT a system will do when it is completed

D) The project's risk assessment report

E) The database schema design

Answer: C) WHAT a system will do when it is completed

Explanation: Slide 5 emphasizes that use cases communicate "WHAT a system will do" but not "HOW a system will do anything."

3. Which of the following is NOT one of the four basic components of a Use Case

Diagram?

A) System

B) Actors

C) Use Cases

D) Relationships

E) Data Flow

Answer: E) Data Flow

Explanation: Slide 6 explicitly lists the four basic components: System, Actors, Use Cases, and Relationships.

4. The rectangle that represents the scope of the system being modeled is called the:

A) Actor Boundary

B) System Boundary

C) Use Case Frame

D) Relationship Link

E) Interaction Box

Answer: B) System Boundary

Explanation: Slide 7 defines the System Boundary as representing "the boundary between the (physical) system and the actors."

5. An Actor in a use case diagram can represent:

A) Only human users

B) Only other software systems

C) Only external hardware devices

D) Roles which may include human users, external hardware, or other systems

E) Only the primary system administrator

Answer: D) Roles which may include human users, external hardware, or other systems

Explanation: Slide 9 defines an actor as a role that "may include human users, external hardware or other systems."

6. A Use Case typically has a name that is a:

A) Noun-only phrase

B) Adjective-Noun phrase

C) Verb-Noun phrase

D) Adverb-Verb phrase

E) Numbered code

Answer: C) Verb-Noun phrase

Explanation: Slide 11 states that a "Use case name typically has a verb-noun phrase."

7. To identify use cases, you should ask which of the following questions? A) What programming language will be used?

B) What are the primary tasks the actor wants the system to perform?

C) How much will the project cost?

D) What is the brand of the server?

E) How many developers are on the team?

Answer: B) What are the primary tasks the actor wants the system to perform?

Explanation: Slide 12 lists this as the first question to ask to identify relevant use cases.

8. The relationship that indicates an actor participates in or communicates with a use case is called a(n):

A) Generalization

B) Include

C) Extend

D) Association

E) Inheritance

Answer: D) Association

Explanation: Slide 15 defines an association as a relationship that "indicates that an actor participates in (i.e. communicates with) the use case."

9. In actor generalization, a child actor:

A) Is completely unrelated to the parent actor

B) Inherits all characteristics and behavior of the parent actor

C) Cannot interact with any use cases the parent interacts with

D) Must have a different name than the parent actor

E) Replaces and eliminates the parent actor

Answer: B) Inherits all characteristics and behavior of the parent actor

Explanation: Slide 17 states that in actor generalization, "A child actor inherits all of the characteristics and behavior of the parent actor."

10. Based on the DVD system example (Slide 18), which actor has the most rights?

A) Attendant

B) Manager

C) Administrator

D) Customer

E) The "Loan DVD" use case

Answer: C) Administrator

Explanation: The Administrator actor is associated with powerful system-level use cases like "Add User" and "Remove User," implying the highest level of rights.

11. The <<include>> **relationship is used to:**

A) Model optional behavior that may occur under certain conditions

B) Avoid describing the same flow of events several times by putting common behavior in a separate use case

C) Show that a child use case inherits from a parent use case

D) Indicate that an actor can communicate with a use case

E) Represent the system boundary

Answer: B) Avoid describing the same flow of events several times by putting common behavior in a separate use case

Explanation: Slide 22 states that the include relationship enables us to "avoid describing the same flow of events several times by putting the common behavior in a use case of its own."

12. In an <<include>> **relationship, the included use case:**

A) Always occurs on its own

B) Never stands alone and only occurs as part of a larger base use case

C) Is optional and may not be executed

D) Overrides the behavior of the base use case

E) Is a more general version of the base use case

Answer: B) Never stands alone and only occurs as part of a larger base use case Explanation: Slide 21 states "The included use case never stands alone. It only occurs as a part of some larger base that includes it."

13. The <<extend>> **relationship is used to model:**

A) Mandatory behavior that always happens

B) Common behavior used in multiple places

C) Optional behavior that occurs under certain conditions

D) The communication between an actor and a use case

E) Inheritance between use cases

Answer: C) Optional behavior that occurs under certain conditions

Explanation: Slide 25 describes the extend relationship where the base use case's behavior "may be extended by the behavior of another use case" under certain conditions, making the extended behavior optional.

14. A key difference between <<include>> **and** <<extend>> **is:**

A) Include is for optional behavior, extend is for mandatory behavior

B) Include is for mandatory behavior, extend is for optional behavior

C) Both are used for exactly the same purpose

D) Include is for actors, extend is for use cases

E) Include uses a dotted line, extend uses a solid line

Answer: B) Include is for mandatory behavior, extend is for optional behavior Explanation: The include relationship is for common, required behavior (mandatory). The extend relationship is for conditional or optional behavior, as shown in Slides 21-26.

15. In the university enrollment example (Slide 26), "Perform Visa check" extends "Enroll in Uni." when:

A) The student is a domestic student

B) The student is any student

C) The student is a foreign student

D) The student has paid tuition

E) The student chooses a major

Answer: C) The student is a foreign student

Explanation: The extension point on Slide 26 is labeled "foreign stu," meaning the "Perform Visa check" behavior is added only if the student is foreign.

16. In a use case generalization relationship:

A) The parent use case inherits behavior from the child

B) The child use case inherits behavior and meaning from the parent

C) The use cases are unrelated

D) It indicates optional behavior

E) It indicates mandatory included behavior

Answer: B) The child use case inherits behavior and meaning from the parent Explanation: Slide 29 states that in use case generalization, "The child use case inherits the behavior and meaning of the parent use case."

17. In the payment example (Slide 30), "Pay via Credit Card" and "Pay via Paypal" are specific examples of the more general use case:

A) Validate User

B) Login

C) Pay

D) Calculate Total

E) Print Receipt

Answer: C) Pay

Explanation: The diagram shows "Pay via Credit Card" and "Pay via Paypal" as specialized children of the general "Pay" use case, demonstrating generalization.

18. According to the relationship summary table (Slide 32), which relationship represents the "insertion of additional behavior into a base use case that does not know about it"?

A) Association

B) Include

C) Extend

D) Use Case Generalization

E) Inheritance

Answer: C) Extend

Explanation: The table on Slide 32 defines the extend relationship with this exact phrasing.

19. What is a Use Case Scenario?

A) A diagram showing the database tables

B) A formal description of the flow of events during the execution of a Use Case

C) The project budget forecast

D) A list of all the actors in the system

E) The code implementation for a use case

Answer: B) A formal description of the flow of events during the execution of a Use Case

Explanation: Slide 34 defines a scenario as a "formal description of the flow of events that occur during the execution of a Use Case instance."

20. The "Main Success Scenario" in a use case specification describes:

A) What happens when everything goes wrong

B) What happens when everything goes to plan (the "happy path")

C) The preconditions that must be true

D) The actors involved

E) The open issues for the use case

Answer: B) What happens when everything goes to plan (the "happy path")

Explanation: Slide 34 states the Main scenario "describes what happens when everything goes to plan" and is also referred to as the "happy path" in the template on Slide 37.

21. Which section of a use case specification describes alternative paths the use case may take, such as error conditions?

A) Preconditions

B) Main Scenario

C) Extensions

D) Postconditions

E) Trigger

Answer: C) Extensions

Explanation: The template on Slide 37 shows that the "Extensions" section is for

"Alternative paths that the use case may take."

22. In the "Withdraw Money" use case example, what is the Trigger?

A) User has logged into ATM

B) User has withdrawn money

C) User has chosen to withdraw money

D) System displays account types

E) System debits the user's account

Answer: C) User has chosen to withdraw money

Explanation: Slide 39 lists the trigger as "User has chosen to withdraw money," which is the action that initiates the use case.

23. In the "Withdraw Money" example, what is the first step of the Main Success Scenario?

A) User enters amount

B) User chooses account type

C) System displays account types

D) System asks for amount to withdraw

E) User removes money

Answer: C) System displays account types

Explanation: The numbered steps on Slide 39 show that Step 1 is "System displays account types."

24. What is a Precondition in a use case specification?

A) The final state of the system after the use case completes

B) The action that starts the use case

C) What needs to be true before the use case can start

D) The list of alternative flows

E) The primary actor's name

Answer: C) What needs to be true before the use case can start

Explanation: The template on Slide 37 defines Preconditions as "What needs to be true before use case 'executes'."

25. The primary purpose of use case modeling in Requirements Engineering is to:

A) Write the source code

B) Act as a means of communicating with stakeholders about what the system is intended to do

C) Design the network infrastructure

D) Hire the development team

E) Calculate the return on investment (ROI)

Answer: B) Act as a means of communicating with stakeholders about what the system is intended to do

Explanation: This is a direct quote from Slide 5, highlighting the communicative role of use cases.

26. An actor is always:

A) Inside the system boundary

B) A part of the system being designed

C) External to the system

D) A use case

E) A relationship

Answer: C) External to the system

Explanation: By definition, actors are external entities that interact with the system, as represented by them being outside the system boundary.

27. The "Validate User" use case in a banking system (Slide 22) is most likely related to "Withdraw Money" and "Transfer Money" by which relationship?

A) Extend

B) Generalization

C) Include

D) Association

E) Realization

Answer: C) Include

Explanation: The diagram on Slide 22 shows "Withdraw money" and "Transfer money" both including "Validate User," indicating the common, mandatory behavior of validating the user is included in those use cases.

28. If a use case "Place Order" must always execute the use case "Calculate Tax," the correct relationship to model this is:

A) Place Order ---> <<extend>> Calculate Tax

B) Place Order ---> <<include>> Calculate Tax

C) Calculate Tax ---> <<extend>> Place Order

D) Calculate Tax ---> <<include>> Place Order

E) Place Order --(Generalization)--> Calculate Tax

Answer: B) Place Order ---> <<include>> **Calculate Tax**

Explanation: Since calculating tax is a mandatory part of placing an order, the include relationship is appropriate. The base use case (Place Order) includes the common use case (Calculate Tax).

29. If a use case "Search Product" can optionally, under the condition "product not found," execute the use case "Suggest Similar Product," the correct relationship is:

A) Search Product ---> <<include>> Suggest Similar Product

B) Search Product ---> <<extend>> Suggest Similar Product

C) Suggest Similar Product ---> <<include>> Search Product

D) Suggest Similar Product ---> <<extend>> Search Product

E) Search Product --(Generalization)--> Suggest Similar Product

Answer: D) Suggest Similar Product ---> <<extend>> **Search Product**

Explanation: The optional behavior ("Suggest Similar Product") extends the base use case ("Search Product") under a specific condition ("product not found"). The arrow points from the extending use case to the base use case.

30. The relationship that shows a "is-a-kind-of" connection between use cases is:

A) Association

B) Include

C) Extend

D) Generalization

E) Dependency

Answer: D) Generalization

Explanation: Generalization represents an inheritance relationship, meaning the child use case "is-a-kind-of" parent use case (e.g., "Pay via Credit Card" is-a-kind-of "Pay").

31. In a Library Management System, which actor would most likely be associated with the "Generate Reports" use case?

A) Guest

B) Student Member

C) Librarian

D) All of the above

E) None of the above

Answer: C) Librarian

Explanation: Based on the example on Slide 20, administrative tasks like "Generate reports" are typically associated with the Librarian or System Admin actor, not general members or guests.

32. What is the correct notation for an Actor in a Use Case Diagram?

A) A rectangle

B) A oval

C) A stick figure

D) A dotted line

E) A solid arrow

Answer: C) A stick figure

Explanation: Slide 9 shows the notation for an actor is a stick figure.

33. What is the correct notation for a Use Case in a Use Case Diagram?

A) A rectangle

B) An oval

C) A stick figure

D) A dotted line E) A solid arrow

Answer: B) An oval

Explanation: Slide 11 shows the notation for a use case is an oval.

34. The line connecting an Actor to a Use Case is called a(n):

A) Inclusion

B) Extension

C) Generalization

D) Association

E) Specialization

Answer: D) Association

Explanation: Slide 15 defines this line as an association, indicating participation/communication.

35. In the "Borrow a book" use case scenario (Slide 42), what is Extension 3a?

A) Login details are invalid

B) User cancels the request

C) The title is for "Reference Only"

D) User has exceeded the borrowing limit

E) System cannot read ISBN

Answer: C) The title is for "Reference Only"

Explanation: The extensions on Slide 42 list 3a as "System notifies the title is for

'Reference Only', exits".

36. The "Postconditions" of a use case describe:

A) The actor's motivation

B) What will be true after the use case successfully executes

C) The steps to follow

D) What must be true before the use case starts

E) The alternative flows

Answer: B) What will be true after the use case successfully executes

Explanation: The template on Slide 37 defines Postconditions as "What will be true after the use case successfully 'executes'."

37. Which of the following is a valid example of an Actor?

A) Login

B) PayPal Payment Gateway

C) Place Order

D) <<include>>

E) System Boundary

Answer: B) PayPal Payment Gateway

Explanation: An actor is an external entity. A payment gateway is an external system that the system being designed would interact with. The others are use cases, relationships, or diagram elements.

38. The "Level" of a use case in its characteristic information typically refers to its:

A) Importance

B) Altitude

C) Abstraction (e.g., user-goal level, sub-function level)

D) Difficulty to implement

E) Number of steps

Answer: C) Abstraction (e.g., user-goal level, sub-function level)

Explanation: While not detailed in the slides, in standard use case templates, "Level" refers to the level of abstraction (e.g., high-level user goal vs. a lower-level subgoal).

39. The purpose of the "Open Issues" section in a use case specification is to:

A) List problems that have been solved

B) Document issues regarding the use case that need resolution

C) Describe the main success scenario

D) List all the actors

E) Define the technical architecture

Answer: B) Document issues regarding the use case that need resolution

Explanation: The template on Slide 37 defines "Open Issues" as "Issues regarding the use case that need resolution."

40. Use Cases are primarily used during which phase of the SDLC?

A) Deployment

B) Requirements Gathering and Specification

C) Implementation

D) Maintenance

E) Testing

Answer: B) Requirements Gathering and Specification

Explanation: The entire lecture positions use cases as a tool for requirement specification (Slide 3, 5), which is part of the early phases of the SDLC.

Diagram-Based & Multiple Answer Questions

41. [MULTIPLE ANS] Which of the following are valid relationships in a Use Case Diagram according to the lecture?

A) Association

B) Include

C) Extend

D) Generalization

E) Implementation

Answer: A) Association, B) Include, C) Extend, D) Generalization

Explanation: Slide 14 and the summary table on Slide 32 list these four as the main types of relationships.

42. [MULTIPLE ANS] Looking at the system diagram on Slide 8, which statements are TRUE?

A) The Actor is outside the System Boundary.

B) The Use Cases are inside the System Boundary.

C) The Actor is associated with the "Login/Logout" Use Case.

D) The "Admin" Actor is inside the System Boundary.

E) The "Place Order" Use Case is associated with the "Inventory Manager" Actor.

Answer: A) The Actor is outside the System Boundary., B) The Use Cases are inside the

System Boundary., C) The Actor is associated with the "Login/Logout" Use Case., E) The "Place Order" Use Case is associated with the "Inventory Manager" Actor. Explanation: The diagram shows actors outside the rectangle (system boundary) and use cases inside. Associations (lines) connect the "Admin" actor to "Login/Logout" and the "Inventory Manager" actor to "Place Order". The "Admin" actor is drawn outside the boundary.

43. [MULTIPLE ANS] From the Library System example on Slide 28, which relationships are depicted?

A) <<include>> between "Borrow book" and "Validate member"

B) <<extend>> between "Search item" and "Suggest add item to library"

C) <<extend>> between "Borrow book" and "Calculate fine"

D) Generalization between "Guest" and "Student member"

E) Association between "Student member" and "Borrow book"

Answer: A) <<include>> **between "Borrow book" and "Validate member",**

B) <<extend>> **between "Search item" and "Suggest add item to library",**

C) <<extend>> **between "Borrow book" and "Calculate fine", E) Association between "Student member" and "Borrow book"**

Explanation: The diagram shows: A <<include>> from "Borrow book" to "Validate member"; A <<extend>> from "Suggest add item" to "Search item" (condition: if not found); A <<extend>> from "Calculate fine" to "Borrow book" (condition: overdue books); An association line between the "Student member" actor and the "Borrow book" use case.

There is no generalization shown between actors in this specific diagram.

44. [DIAGRAM] Based on the description in Activity 06 (Slide 31), which use case relationship would likely exist between "Search" and "Show enrollment requirements"?

A) Search <<include>> Show enrollment requirements

B) Search <<extend>> Show enrollment requirements

C) Show enrollment requirements <<include>> Search

D) Show enrollment requirements <<extend>> Search

E) Search --(Generalization)--> Show enrollment requirements

Answer: D) Show enrollment requirements <<extend>> **Search**

Explanation: The activity states "if the user needs, the system will show the enrollment requirements". This describes optional behavior that extends the base "Search" functionality under a condition ("if the user needs"), making extend the correct relationship. The arrow points from the extending use case to the base.

45. [DIAGRAM] In the "Pay" generalization example (Slide 30), what does the relationship imply?

A) You must use all three payment methods every time.

B) "Pay" is an abstract use case that is concretely implemented by its children.

C) "Pay via Credit Card" includes the behavior of "Pay via PayPal".

D) "Pay" is optional and extended by the other use cases.

E) The actor only interacts with the specific payment methods, not the general "Pay" use case.

Answer: B) "Pay" is an abstract use case that is concretely implemented by its children.

Explanation: Generalization implies that the parent ("Pay") is a general classification. The specific ways to pay (via Credit Card, PayPal, EFT) inherit the general behavior and meaning of "Pay" and add their own specific details. The actor would likely be associated with the general "Pay"